

Shubham Mahindrakar

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https://github.com/ShubhamBioIT

PROFILE

Dynamic AI-Bioinformatics Engineer combining deep learning expertise with bioinformatics analysis and full-stack tool development (Docker, Streamlit). Skilled in Python, R & web development. Building advanced multimodal platforms (ONCO-JEPA) to automate complex multi-omics analyses and deliver high-impact clinical insights.

TECHNICAL SKILLS

- AI & Machine Learning:** Deep Learning (CNNs, Transformers), Generative AI, Agentic AI, LLMs, Scikit-Learn.
- Bioinformatics & Genomics:** Single-Cell RNA-seq, Multi-omics Integration, Pseudobulk Analysis, Genome Annotation, Molecular Docking.
- Programming & Data Science:** Python, R, SQL (MySQL), Pandas, NumPy, Data Visualization.
- DevOps & Tool Development:** Snakemake, Docker, Conda, Linux/Bash, Streamlit, HTML5/CSS, Git/GitHub.

PROFESSIONAL EXPERIENCE

- AI-Bioinformatics Research Intern** 01-2026 – Present
EFREI Lab (Remote, Paris, France)
- Architecting the **ONCO-JEPA Platform** for NSCLC, deploying multimodal deep learning models on multi-omics data via Docker, **Federated Learning**, and **Agentic AI**.
 - Executed advanced single-cell to pseudobulk RNA-seq transformations to extract high-confidence differential expression profiles.
 - Engineered automated Python visualization pipelines to accelerate clinical interpretation of metagenomic and Antimicrobial Resistance (AMR) data.
- Bioinformatics Developer Intern** 11-2024 – 07-2025
Amrita Mind Brain Centre, Kerala
- Engineered an interactive web-based simulation tool for RNA secondary structure prediction (Zuker's Algorithm) utilizing HTML5, CSS, and JavaScript.
 - Delivered this platform as a core component for the **Government of India's Virtual Labs initiative** to build scalable, user-facing scientific tools.
- Machine Learning Intern** 03-2025 – 05-2025
Ediglobe (Remote, Bangalore)
- Developed and trained a machine learning classification model to predict the 10-year risk of Coronary Heart Disease (CHD) utilizing large-scale clinical datasets.
 - Applied advanced Python data science libraries for data preprocessing, normalization, and predictive modeling.

PROJECTS

- Advanced Transcriptomic Resolution via Pseudobulk RNA-seq Integration,** 04-2026 – Present
EFREI Lab (Paris)
Executed advanced single-cell to pseudobulk RNA-seq data transformations to accurately resolve complex expression profiles and extract high-confidence differential gene markers/drivers using R programming.
- Automated Molecular Cloning Sequence Optimizer (Mutate-for-Digest)** 03-2026 – Present
Built a full-stack Python/Streamlit web application that automates the introduction of silent mutations for molecular cloning, accelerating DNA sequence optimization for laboratory research teams.
- Scalable scRNA-seq Automation & Transcriptomics Workflow,** 10-2025 – 12-2025
Amrita School of Biotechnology
Deployed an end-to-end automated pipeline (integrating Cell Ranger for upstream and custom scripts for downstream analysis), established a scalable foundation for multi-omics research.
- Automated Prokaryotic Genome Annotation Pipeline,** *Amrita School of Biotechnology* 08-2025 – 12-2025
Engineered a fully automated, Docker/Conda-containerized workflow integrating 7+ bioinformatics tools (Prokka, MEME Suite). Enabled high-throughput batch processing with interactive HTML reporting for the Bharat Genome Database.

CRISPR/Cas9 Off-Target Prediction via Deep Learning , <i>Amrita School of Biotechnology</i> Developed custom CNN and multi-scale neural network architectures to predict off-target effects. Successfully handled 96.8% class imbalance via stratified sampling, achieving a 25% AUPRC improvement and 99.3% recall.	08-2025 – 11-2025
Interactive Bioactivity & Drug-Target Binding Dashboard , <i>Amrita School of Biotechnology</i> Engineered an interactive Streamlit platform to automate the preprocessing, normalization, and visualization of complex bioactivity data (Ki values), accelerating drug-target binding analysis for early-stage drug discovery.	03-2025 – 05-2025

EDUCATION

M.Sc. Bioinformatics , <i>Amrita School Of Biotechnology</i> Current CGPA - 8.41	08-2024 – present Kollam, India
B.Sc. (Hons.) Biotechnology , <i>Walchand Centre For Biotechnology</i> CGPA - 9.06	08-2020 – 07-2023 Solapur, India

COURSES

Large Language Models (LLMs) & AI Agents , <i>Simplilearn</i>	01-2026 – 02-2026
Deep Learning with Python: Sequence Models and Transformers , <i>LinkedIn Learning</i>	10-2025 – 10-2025
Deep Learning and Generative AI: Data Prep, Analysis, and Visualization with Python , <i>LinkedIn Learning</i>	10-2025 – 10-2025
Python for Data Science, AI & Development , <i>Coursera</i>	01-2025 – 03-2025

AWARDS

Best Paper Award (First Place) - ICSRF'2025 Awarded First Place for the research paper titled 'Self-Guided Bioinformatics Learning Through Virtual Labs: Towards Sustainable, Inclusive Learning' presented at ICSRF 2025, Amritapuri campus, Kerala, India.	01-09-2025
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PUBLICATIONS

Artificial Intelligence in Vaccine Development: Applications, Implementation, and Future Directions (Review article) Submitted (Under Review)	04-2026
Self-Guided Bioinformatics Learning Through Virtual Labs: Towards Sustainable, Inclusive Learning , <i>Shubham Satish Mahindrakar, Dhanush Kumar, Rakhi Radhamani, Asha Vijayan, Krishnashree Achuthan, Shyam Diwakar</i> https://dx.doi.org/10.2139/ssrn.5787882  SSRN (Social Science Research Network)	30-08-2025

CONFERENCE

ICSRF'2025 , <i>Amrita Vishwa Vidyapeetham, Amritapuri, Kerala</i> Presented award-winning research on sustainable, AI-guided virtual lab ecosystems.	08-2025 – 09-2025
Aavishkar'2022- Project Competition , <i>Solapur University, Maharashtra</i> Presented undergraduate research on sustainable biowaste management.	12-2022 – 12-2022

WORKSHOPS

- Machine Learning in Genomics-IBAB 
- National Seminar and Workshop on Next-Generation Sequencing & Data Analysis 
- Molecular Dynamics Simulation 

ORGANISATIONS

Bioclues Organization (India's Bioinformatics Society) , <i>Member</i>	06-2024 – Present
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LANGUAGE PROFICIENCY

English, Hindi, Marathi Read, Write, Speak and Understand	Telugu , Kannada Understand
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